

Tulips Project: Component Datasheets and References

The Tulips monitoring system is built using several hardware components that work together to collect environmental data, display information, detect unsafe conditions, and control external devices. The datasheets and product references listed below provide detailed technical specifications and usage information for each component.

Item	Component	Datasheet / Reference	Description
a	1.3" White I2C 128×64 OLED Display	  1.3 Inch I2C OLED Display Module PDF - Bibuke / SH1106G	Datasheet for the 1.3-inch OLED display module commonly used to show sensor readings and system status. The display operates using the SH1106G controller. Many OLED displays use either the SSD1306 or SH1106 controller.
a	OLED Controller Reference	 SSD1306 Datasheet - Solomon Systech, via Adafruit	This reference helps developers understand display communication and configuration.
b	ESP32S DevKit Wi-Fi + BLE (30-pin)	 DOIT ESP32 DEVKIT V1 30-pin board reference	Product reference for the widely used ESP32 development board that serves as the project's main controller.
b	ESP32 Module Reference	 ESP32-WROOM-32 Datasheet - Espressif	Official technical documentation for the ESP32-WROOM-32 module, covering wireless communication, processing capabilities, and electrical characteristics.

Item	Component	Datasheet / Reference	Description
c	DHT22 / AM2302 Temperature and Humidity Sensor	 DHT22 Datasheet PDF - Aosong, via MikroE	Provides operating specifications, accuracy ratings, and communication protocols for temperature and humidity measurement.
c	DHT22 / AM2302 Alternative Reference	 AM2302 Datasheet - DigiKey	Additional technical reference for the same sensor under its alternative product name, AM2302.
d	MQ-5 Gas Sensor	 MQ-5 Datasheet - Hanwei Electronics	Datasheet for detecting LPG, natural gas, and coal gas concentrations in the environment.
d	MQ-5 Sensor Module Reference	Natural Gas, LPG Sensor MQ5 Breakout Datasheet - DigiKey PDF	Useful when working with the commonly available MQ-5 sensor module rather than the standalone sensing element.
e	5V Single-Channel Relay Module	 SRD-05VDC-SL-C Relay Datasheet - Songle, via AllDataSheet	Technical specifications for the Songle relay used in many 5V relay modules.

Item	Component	Datasheet / Reference	Description
e	Relay Module Reference	Relay module documentation - Soldered	Product documentation covering wiring, operation, and safety considerations for relay modules.

Project Application Summary

In the Tulips project, the ESP32S DevKit acts as the central processing unit, coordinating data collection, display updates and control operations. The DHT22 (AM2302) sensor continuously measures temperature and humidity levels around the tulips, helping maintain optimal growing conditions. These readings are displayed in real time on the 1.3-inch OLED display, allowing users to monitor environmental conditions easily.

To enhance safety, the MQ-5 gas sensor is used to detect the presence of gases such as LPG, natural gas or coal gas that may pose a risk to the surrounding environment. When predefined conditions are met, the 5V relay module can automatically switch connected devices such as fans, water pumps, lighting systems, or alarm units. This combination of sensing, monitoring, and automated control creates a smart and efficient environmental monitoring solution for the tulip-growing setup.